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SECTION 52

Report: Cybersecurity Threats in the Digital Age

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1.0 INTRODUCTION

In the era of digital transformation, the global reliance on Internet of Things (IoT) devices has grown exponentially. This situation has indirectly caused a huge concern called cybersecurity threats. A cybersecurity threat is any potential harmful events that can negatively impact digital systems, networks and data. While enjoying the convenience and perks brought by the Internet, individuals and organizations are also exposed to many kinds of unseen dangers. According to recent reports, the frequency of cyber attacks are on the rise, with phishing and malware emerging as two of the most pervasive threats faced today. (Wheeler, 2024) In response to the problem, this report investigates the fundamental causes and effects of prevalent cybersecurity threats, as well as the possible solutions of it.

2.0 BACKGROUND OF THE PROBLEM

This part discussed the cause and effects of the problem.

2.1 Causes of the Problem

Cybersecurity threats sprout from a combination of technical vulnerabilities and human factors. One of the significant causes is **human error**. Internet users might unintentionally download malicious software online, wrapped and disguised as normal useful software, or fall for phishing scams, which trick them into revealing sensitive information to unauthorized personnel. Phishing attacks over text messages or even phone calls are particularly effective now with the technology of replicating personalities and voices of a trusted person using artificial intelligence, while also exploiting human psychology, for example fear of their family being hurt. Other examples include phishing emails that appear to be from a legitimate source and are delivered to the inbox, making them hard to detect.

Besides, the **vulnerabilities of security measures** are also a primary cause of increasing cybersecurity threats. This includes software flaws, outdated operating systems, and misconfigured security settings, which provide a perfect entry point for attackers to exploit.

Cybercriminals are known to target these weaknesses to deploy viruses and malware, as they are not protected by any security updates, fixes and patches. This is compounded by a lack of security measures, such as failing to implement firewalls or educate people and employees on security practices, thus leaving organizations prone to attacks. A survey on people working in the IT field shows that less than 50% of IT workers aside from those working for the Security Operation Centre had the techniques of handling most of the cybersecurity attacks, highlighting the lack of skills of most employees on handling cybersecurity attacks (Mohannadi et al., 2018).

Furthermore, the **lack of awareness among the community** also leads to the increase of cybersecurity threats. When people are unaware of the threat, they cannot protect themselves against it. The ignorance behaviour causes them to fall easily for online scams, viruses and malwares. Without proper education of cybersecurity, 67% workers are concerned for lacking fundamental security awareness (Alger & Denyer, 2024) as they might unknowingly put themselves, or the entire company at risk by engaging in risky online behaviour or fail to recognise suspicious activity.

2.2 Effects of the Problem

The consequences of a successful cyber attack is catastrophic for both individuals and organizations. The most immediate effect is often **significant financial loss**. This can result from scams, theft of fundings, the cost of system maintenance, paying ransoms in ransomware attacks, and regulatory fines. For example, the ransomware attack on Change Healthcare in 2024 is estimated to have cost its parent company, UnitedHealth Group, approximately \$2.87 billion. (CyberManagementAlliance, 2025) Another news in Petaling Jaya stated that a 52-year-old man lost nearly a million to phishing scams, including the personal and company funding of his. (FMT Reporters, 2025) Meanwhile in the United States, a father suicided after losing his entire savings to 'pig butchering' crypto scam (Rebane & Watson, 2024), a type of online scam where the victim is encouraged to make increasing financial contributions over a long period. (Wikipedia, n.d.)

Furthermore, data breaches lead to the **theft of sensitive information**, including personal data, financial records, and intellectual property. The loss of this data can result in identity theft for individuals. Personal identifying information such as full name, home address and birth date can be used for illicit schemes such as filing a fraudulent tax return and making up numbers or information to guarantee they get a large refund (DeNicola, 2025). In short, the stolen personal information may be exploited to impersonate individuals, collect government benefits, drain funds, change account credentials, add unauthorized users, or gain access to medical services, resulting in financial loss and inaccurate records (DeNicola, 2025).

Finally, cyber attacks can cause major **disruption of services**. A well schemed distributed denial-of-service (DDoS) attack can pose a danger to well known servers such as AWS, Google and Cloudflare (A10, 2024) by overwhelming the target or its surrounding infrastructure with a flood of Internet traffic (Cloudflare, n.d.). The attack can result in websites or applications becoming completely inaccessible to legitimate users, leading to significant downtime, loss of revenue and damaging the reputation of the affected organizations. Moreover, the disruption to critical online services such as e-commerce platforms will affect millions of users and affect businesses that rely on these platforms for their daily operations.

3.0 POSSIBLE SOLUTIONS

This part describes three possible solutions to the problem.

3.1 Raising Awareness of Cybersecurity Threats

One of the solutions to the problem is to raise awareness of the importance of cybersecurity among society, starting from students to employees. This topic should be encapsulated in textbooks, stories, assignments or forums in a variety of subjects such as languages, moral education and computer science. Primary and secondary schools should have awareness talks or seminars about cybersecurity as co-curricular activities once in a while, not only targeted towards students but also encouraging everyone else to participate. Companies

should invest in educating employees and make sure all of them have basic knowledge about what cyberattacks are and how to identify and prevent them. The official government unit should try to spread awareness as much as possible. They should post creative posters and videos about the threats and effects of cyberattacks on social media to attract the attention of people. Elderly people shouldn't be left out either. Copies of posters and booklets should also be handed out at places such as markets and public parks, targeted towards the elderly who may not be familiar with social media. Having basic acknowledgement of the usual patterns of cybersecurity attacks reduces the risk of falling into or getting affected by it generally. Unavoidably, as these practices are being implemented for a prolonged period, people might think it becomes repetitive and monotonous, as they are so used to the knowledge all the time.

3.2 Supervising the Underage

Parents and guardians should only let their children and adolescents use the internet under supervision, while not exposing the internet to toddlers and young children. This is because the underaged do not have a fully developed mindset yet and might not be able to distinguish between scams and genuine relationships. Meanwhile, the supervision of parents when they are using the internet for activities such as learning or connecting with others can prevent them from malicious activities, attacks and the influence of cyber threats. Supervision of parents should include limiting the screen time of their children, applying parental control on social media and video games and reviewing any form of transaction before they have been made. Parents should also set a safety word with their children so that they can distinguish between scam calls intimidating their child and real danger. By having parents supervising their online activities, they can protect the minors from online scams, predators and inappropriate content exposure. Surprisingly, research has shown that excessive parental supervision practices can lead to problematic behaviour of children, such as losing the development of adolescence autonomy and self regulations (Wisniewski et al., 2017).

4.0 CONCLUSION

In conclusion, the growing dependence on digital technology and Internet of Things (IoT) devices has brought great convenience but also increased the risk of cybersecurity threats. This report suggested two solutions to the problem, which are raising the awareness among the community and supervision of minor online activity by parents or guardians. These measures are expected to reduce the likelihood of individuals falling victim to scams and online fraud, as well as preventing any potential losses by enabling early detection of suspicious activities of cyber attacks. With greater awareness and responsible actions from all parties, a safer and more secure digital environment can be achieved for everyone.

REFERENCES

- A10. (2024, May 30). Five Most Famous DDoS Attacks and Then Some. *A10 Networks*.
<https://www.a10networks.com/blog/5-most-famous-ddos-attacks/>
- Alger, J., & Denyer, C. (2024, October 24). 67% of organizations say employees lack basic security awareness. *Security Magazine*.
<https://www.securitymagazine.com/articles/101154-67-of-organizations-say-employees-lack-basic-security-awareness>
- Cloudflare. (n.d.). *What is a distributed denial-of-service (DDoS) attack?* Cloudflare. Retrieved July 1, 2025, from <https://www.cloudflare.com/learning/ddos/what-is-a-ddos-attack/>
- CyberManagementAlliance. (2025, January 20). Top 10 Biggest Cyber Attacks of 2024 & 25 Other Attacks to Know About! *Cyber Management Alliance*.
<https://www.cm-alliance.com/cybersecurity-blog/top-10-biggest-cyber-attacks-of-2024-25-other-attacks-to-know-about>
- DeNicola, L. (2025, April 8). What Can Identity Thieves Do with Your Personal Info and How Can You Protect Yourself? *Experian*.
<https://www.experian.com/blogs/ask-experian/what-can-identity-thieves-do-with-your-personal-information-and-how-can-you-protect-yourself/>
- FMT Reporters. (2025, June 28). Businessman loses nearly RM1mil in tender scam. *Free Malaysia Today*.
<https://www.freemalaysiatoday.com/category/nation/2025/06/28/businessman-loses-nearly-rm1mil-in-tender-scam>

Mohannadi, A., Awan, H., Hamar, I., Hamar, J., Shah, Y., Musa, M., & Ahmad. (2018).

Understanding Awareness of Cyber Security Threat among IT Employees.

https://www.researchgate.net/publication/328241670_Understanding_Awareness_of_Cyber_Security_Threat_among_IT_Employees

Rebane, T., & Watson, I. (2024, June 20). Pig Butchering Scam. *Killed by a scam: A father took his life after losing his savings to international criminal gangs. He's not the only one.*

<https://edition.cnn.com/2024/06/17/asia/pig-butchering-scam-southeast-asia-dst-intl-hnk>

Wheeler, K. (2024, September 25). Top 10: Biggest Cyber Threats.

<https://cybermagazine.com/top10/top-10-biggest-cyber-threats>

Wikipedia. (n.d.). Pig butchering scam - Wikipedia. Retrieved July 1, 2025, from

https://en.wikipedia.org/wiki/Pig_butchering_scam

Wisniewski, P., Ghosh, A. K., Xu, H., Rosson, M. B., & Carroll, J. M. (2017, February 27).

Parental Control vs. Teen Self-Regulation: Is there a middle ground for mobile online safety? 10.1145/2998181.2998352

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